Cylinders, capacity and missing heights

Day 19 • 2026-12-01 • 2:15 PM–3:30 PM

15 questions. Show your method and include units. For multiple choice, enter A, B, C or D. Follow your instructor’s calculator guidance.

Name _____ **Date** _____

Type into the fields and save a copy, or print and write.

01 READ & REASON

Which formula finds a cylinder’s volume when radius r and height h are known?

A. $V = 2 \times \pi \times r \times h$

B. $V = \pi \times r \times h$

C. $V = \pi \times (2r)^2 \times h$

D. $V = \pi \times r^2 \times h$

Choice (A–D) _____

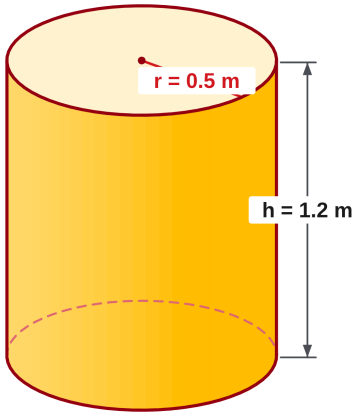
Working / reason _____

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

Cylinders, capacity and missing heights

Day 19 • 2026-12-01 • 15 questions

02 The Shape Canvas cylinder has radius 0.5 m and height 1.2 m. Find volume using pi = 3.14.



Shape Canvas

Answer (m³)

Working / reason

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

Cylinders, capacity and missing heights

Day 19 • 2026-12-01 • 15 questions

03 When using Numeracy Hub to calculate a cylinder from diameter, what must be done before squaring?

A. Subtract the height

B. Divide diameter by 2 to get radius

C. Double diameter

D. Square the diameter and use it as r^2



Numeracy Hub

Choice (A–D)

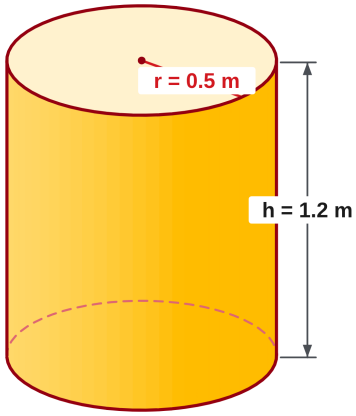
Working / reason

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

Cylinders, capacity and missing heights

Day 19 • 2026-12-01 • 15 questions

04 Use the Shape Canvas cylinder’s volume of 0.942 m³. Express it in litres.



Shape Canvas

Answer (L)

Working / reason

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

05 Which rearrangement finds cylinder height from volume V and radius r?

A. $h = V \times pi \times r^2$

B. $h = pi \times r^2 \div V$

C. $h = V \div (2 \times pi \times r)$

D. $h = V \div (pi \times r^2)$

Choice (A–D)

Working / reason

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

Cylinders, capacity and missing heights

Day 19 • 2026-12-01 • 15 questions

06 CALCULATE

A cylinder has radius 0.3 m and height 2 m. Find its volume using pi = 3.14.

Answer (m³) | **Working / reason**

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

07 A tank has internal diameter 1 m and height 1.5 m. Find its capacity in litres using pi = 3.14.

Answer (L) | **Working / reason**

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

08 A cylinder has volume 3.14 m³ and radius 0.5 m. Find its height using pi = 3.14.

Answer (m) | **Working / reason**

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

09 A cylindrical tank contains 628 L and has radius 0.4 m. Find the liquid depth using pi = 3.14.

Answer (m) | **Working / reason**

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

Cylinders, capacity and missing heights

Day 19 • 2026-12-01 • 15 questions

10 Use 1 US gal = 3.78541 L. Convert 75 L to US gallons, rounding to 2 decimal places.

Answer (US gal)

Working / reason

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

11 APPLY & CHECK

A student enters V/pi × r² to find h. Why should brackets be used?

A. Only V needs to be squared

B. Multiplication should happen after division by pi

C. The radius must be added to height

D. The entire product pi × r² must divide V

Choice (A–D)

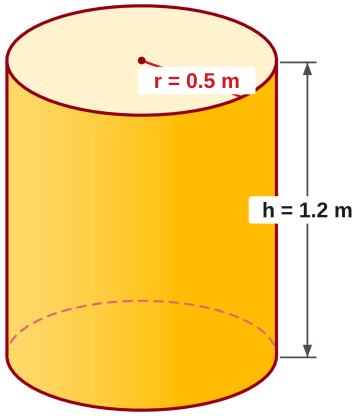
Working / reason

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

Cylinders, capacity and missing heights

Day 19 • 2026-12-01 • 15 questions

12 The Shape Canvas cylinder holds 942 L using pi = 3.14. It is half full. How many litres does it contain?



Shape Canvas

Answer (L)

Working / reason

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

13 A height from V/(pi r²) is to be in metres. Which input units are consistent?

A. V in m³ and r in mm without conversion

B. V in m³ and r in m

C. V in L and r in mm without conversion

D. V in m² and r in m

Choice (A–D)

Working / reason

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

Cylinders, capacity and missing heights

Day 19 • 2026-12-01 • 15 questions

14 A rectangular tank with internal dimensions 1.2 m by 0.8 m needs to hold 480 L. Find required liquid depth.

Answer (m)

Working / reason

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

15 Why must a gallon conversion name US or Imperial gallons?

A. Only Imperial gallons measure liquids

B. Only US gallons can be converted

C. A gallon always equals exactly 4 L

D. The two gallon units have different volumes

Choice (A–D)

Working / reason

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203